

Supplement to: What Does a Sim-to-Real Correlation Support?

Tri Lam

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S1. Consensus and bounded-sample facts

fact	result	interpretation
Included papers	26	non-systematic claim-based corpus
Papers with recovered numeric results	19	recovery from published figures or tables
Finite displayed panel	26/26	all audited papers
Defined target population or probability-sampling mechanism	0/26	none reported
Coefficient p-value	5/26	null-test output, not interval uncertainty in magnitude
Coefficient interval	1/26	interval uncertainty for coefficient magnitude
Neither p-value nor coefficient interval	20/26	observable reporting fact; not evidence the coefficient is wrong

S2. Unit-count sensitivity

The two columns answer a policy/checkpoint sensitivity question; they do not treat tasks or conditions as automatically exchangeable units.

paper	legacy policy/lineage blocks	permissive checkpoints/variants
real2sim-eval	4	20
RoboWorld	8	8
Digital Cousins	4	4
SIMPLER	4	6
SimFoundry	5	8
WorldGym	3	3
RoboSnap	1	10
REALM	3	3
PolaRiS	4	4
SC3-Eval	1	7

paper	legacy policy/lineage blocks	permissive checkpoints/variants
WorldEval	4	4
A Practical Recipe	5	5
Cosmos-Surg-dVRK	3	6
Gemini/Veo	8	8
DreamDojo	1	6
dWorldEval	1	5
WEAVER	2	2
PlayWorld	18	18
EmbodiedSplat	2	4
MolmoSpaces	3	8
Mem-World	2	2
Colosseum V2	1	1
VISER	1	2
OSCAR	2	7
Hi-WM	2	4
WM-PolicyEval	3	3

Across these two explicit codings, **23–25 of 26** papers have fewer than ten policy/checkpoint blocks.

S3. Axis-specific best-case permutation resolution

These are combinatorial resolutions conditional on valid exchangeability, not exact test results. Tied statistics can make the attained minimum coarser.

paper	printed p	policy k	1/k!	task k	1/k!
RoboWorld	<0.001	8	0.000025	—	—
WorldGym	<0.001	3	0.166667	17	0.000000
REALM	<0.001	3	0.166667	7	0.000198
Cosmos-Surg-dVRK	<0.001	3	0.166667	4	0.041667
Mem-World	<0.001	2	0.500000	5	0.008333

S4. Complete leave-one-unit/point table

The 0.10 column is retained only as a descriptive screen. The continuous movement and deletion unit are the reported quantities.

dataset	points	deletion units	unit	r	max abs change in r	max abs change in rho	>0.10
real2sim-eval	17	4	policy	0.9444	0.006	0.054	no
toy							

dataset	points	deletion units	unit	r	max abs change in r	max abs change in rho	>0.10
real2sim-eval rope	20	4	policy	0.9007	0.029	0.073	no
real2sim-eval T-block	15	4	policy	0.9147	0.048	0.090	no
RoboWorld 10a	8	8	policy	0.9005	0.075	0.042	no
RoboWorld 10b	8	8	policy	0.8762	0.169	0.113	yes
RoboWorld 9a	8	8	policy	0.9888	0.191	0.021	yes
RoboWorld 9b	8	8	policy	0.9438	0.165	0.090	yes
Digital Cousins (unit = policy)	16	4	policy	0.9094	0.013	0.047	no
Digital Cousins (unit = gen. level)	16	4	generalization level	0.9094	0.037	0.063	no
Cosmos-Surg- dVRK automated	24	3	policy run	0.7561	0.073	0.086	no
Cosmos-Surg- dVRK manual	24	3	policy run	0.7180	0.123	0.131	yes
DreamDojo	6	6	point	0.9953	0.010	0.057	no
MolmoSpaces pick	8	8	point	0.9585	0.020	0.000	no
MolmoSpaces open	4	4	point	0.8523	0.617	0.900	yes
MolmoSpaces close	4	4	point	0.9727	0.040	0.000	no
REALM Overall	21	3	policy	0.9187	0.025	0.022	no
REALM Default	21	3	policy	0.8838	0.135	0.057	yes
REALM VB-POSE	21	3	policy	0.9335	0.080	0.187	no
REALM V-VIEW	14	3	policy	0.8902	0.168	0.080	yes

dataset	points	deletion units	unit	r	max abs change in r	max abs change in rho	>0.10
subject paper toy, 200 episodes	17	4	policy	0.8970	0.011	0.095	no
subject paper rope, 200 episodes	20	4	policy	0.9181	0.043	0.088	no
subject paper T-block, 200 episodes	15	4	policy	0.9501	0.037	0.071	no
VISER Octo	4	4	point	0.9988	0.000	0.028	no
VISER	5	5	point	0.8496	0.135	0.300	yes
OpenVLA							
OSCAR	7	7	point	0.8552	0.086	0.132	no
Hi-WM	12	12	point	0.9540	0.014	0.019	no
WM- PolicyEval	12	12	point	0.7193	0.260	0.130	yes
Cosmos							
WM- PolicyEval	12	12	point	0.2772	0.176	0.237	yes
IRASim							

Median maximum absolute change in $r = 0.061$; range 0.000–0.617. The descriptive 0.10 screen selects 10/28 illustrative rows; this is not a corpus-prevalence estimate.

S5. Bounded decision cases

Each row concerns only the displayed policies and the stated aggregation rule. Top-1 agreement means that the simulator-selected winner belongs to the real-data winner set; regret is the displayed real-success gap to the best policy.

case	declared rule	r	top-1 result	real regret	diagnostic
SIMPLER Google	equal-weight mean over 5 displayed tasks	0.974	agreement	0	3/5 individual tasks agree; 5/5 leave-one-task-out aggregates agree
SIMPLER WidowX	equal-weight mean over 4 displayed tasks	0.950	agreement	0	3/4 individual tasks agree; 4/4 leave-one-task-out aggregates agree

case	declared rule	r	top-1 result	real regret	diagnostic
Real2Sim T-block	select each policy's best simulated checkpoint	0.878–0.947	disagreement	12.50 pp	non-winning checkpoint tie changes r but not the selected or real winner
A Practical Recipe	printed top-ranked variant in each rank panel	varies	3/11 panels disagree	—	largest printed Pearson r among disagreements = 0.672

Across Real2Sim’s three tasks and three declared checkpoint-collapse rules, 7/9 cells necessarily disagree, 1/9 robustly agrees, and 1/9 is tie-dependent. These nine cells share data and are not independent repetitions.

The vector extraction contains two coincident rope/DP markers at (0,0). Removing one changes all-checkpoint r from 0.901 to 0.881 and mean-rule r from 0.697 to 0.538, but it does not change any of the three rope rule-level top-1 conclusions.

S6. Complete direct-cell matrices

All eligible direct-cell matrices derived from the retained source inventory under the declared candidate/block rules; outcome-exposed and not a prevalence denominator.

panel	source metric bundle	audit Pearson r	audit Spearman rho	top-1	regret (pp)	LOTO
Cosmos-Surg-dVRK — automated fig1b	Pearson + MMRV	0.941	0.829	correct	0.00	1/4
Cosmos-Surg-dVRK — manual human vs dvrk	Pearson + MMRV	0.883	0.371	wrong	10.00	0/4
Digital Cousins	Pearson	0.997	1.000	correct	0.00	4/4
EmbodiedSplat	Pearson	0.936	0.800	correct	0.00	2/2
—						
mesh-conditions						
Hi-WM	Pearson	0.964	1.000	correct	0.00	3/3
Mem-World	Pearson + p-value	1.000	1.000	correct	0.00	5/5
MolmoSpaces — common-appendix-roster	Pearson + Spearman	0.982	0.800	correct	0.00	3/3

panel	source metric bundle	audit Pearson r	audit Spearman rho	top-1	regret (pp)	LOTO
REALM — Default	Pearson + MMRV	1.000	1.000	correct	0.00	7/7
REALM — Overall	Pearson + MMRV	1.000	1.000	correct	0.00	7/7
REALM — VB-POSE	Pearson + MMRV	1.000	1.000	correct	0.00	3/7
SIMPLER — google robot	Pearson + MMRV	0.974	1.000	correct	0.00	5/5
SIMPLER — widowx	Pearson + MMRV	0.950	1.000	correct	0.00	4/4
WEAVER — CtrlWorld	Pearson + Spearman	1.000	1.000	correct	0.00	5/5
WEAVER — WEAVER	Pearson + Spearman	1.000	1.000	correct	0.00	5/5
WEAVER — WEAVER-FT	Pearson + Spearman	1.000	1.000	correct	0.00	5/5
WM-PolicyEval — Cosmos	Pearson + MMRV	0.975	1.000	correct	0.00	4/4
WM-PolicyEval — IRASim	Pearson + MMRV	0.507	0.500	wrong	27.50	1/4
WorldEval	Pearson	0.996	0.800	correct	0.00	5/5
WorldGym	Pearson + ranking- preservation analysis	0.992	1.000	correct	0.00	17/17

The complete-matrix set contains 17/19 displayed top-1 agreements and 2/19 displayed top-1 disagreements. These counts describe this non-systematic recovered set only.

S7. Illustrative cross-source decision atlas

Heterogeneous finite-panel cases selected for evidentiary roles; not a calibration sample, prevalence denominator, or common population.

case	r	simulated winner	displayed real winner	regret (pp)	LOTO	evidence
WorldGym	0.992	OpenVLA	OpenVLA	0.00	17/17	exact published table
Digital Cousins	0.997	pi_0	pi_0	0.00	4/4	table-validated recovered values

case	r	simulated winner	displayed real winner	regret (pp)	LOTO	evidence
SIMPLER	0.974	rt-1-converged	rt-1-converged	0.00	5/5	official source arrays
Google Real2Sim T	0.878–0.947	pi0	dp	12.50	—	vector-PDF recovery; printed coefficient reproduced
best-sim checkpoint						
OSCAR Skeleton	0.855	PG-FAST+	pi0-FAST	1.10	—	printed one-decimal bar labels; r reproduced to rounding
Cosmos-Surg manual	0.883	GR00T N1.5 50k	GR00T N1 20k	10.00	0/4	vector-PDF recovery; pooled coefficient reproduced
WM-PolicyEval / Cosmos	0.975	OpenVLA	OpenVLA	0.00	4/4	vector-PDF recovery plus exact appendix real values
WM-PolicyEval / IRASim	0.507	Octo-Base	OpenVLA	27.50	1/4	vector-PDF recovery plus exact appendix real values

S8. Finite-real-trial remeasurement

Model-conditional real-trial measurement uncertainty for cases with documented real denominators; simulator values held fixed where simulator rollout counts are unavailable. Candidate/block probabilities are independent in the fitted model; cross-candidate coupling is not identified by published marginal counts.

case	posterior P(sim winner is real-best)	posterior expected regret (pp)	real denominator	simulator denominator
Cosmos-Surg manual	0.087508	11.64	10 trials per policy-task	10 initial states x 3 generated seeds x 2 raters; aggregate allocation/dependence unreleased
WM-PolicyEval / Cosmos	0.999852	0.00	20 trials per policy-task	unstated
WM-PolicyEval / IRASim	0.000168	24.99	20 trials per policy-task	unstated

S8.1 Missing simulator-evidence sensitivity

The effective simulator evidence grid is a sensitivity design, not an estimate of the unreleased rollout denominator.

evaluator	sampled-winner match range	expected real-regret range (pp)	all scenarios below one half?	first listed crossing by prior (.5/1/2)
Cosmos	0.3333–0.9999	0.0–18.6	false	2/2/5
IRASim	0.0001–0.4320	13.2–26.2	true	none/none/none

S8.2 Policy-heterogeneous simulator evidence

The sampled-winner estimand measures latent-rank concordance under one common assumed evidence size, not observed-action reliability. The next rows vary assumed evidence by policy; actual evidence remains unknown.

IRASim scenario	evidence (Octo-Base/Octo-Small/OpenVLA)	latent-winner concordance	MCSE	posterior-mean simulator winner
common_10	10/10/10	0.4302	0.00070	Octo-Base
openvla_10	500/500/10	0.7339	0.00062	OpenVLA
openvla_0	500/500/0	0.9821	0.00019	OpenVLA

S8.3 Probability-level affine calibration

Cell-rate MSE and the empirical individual-outcome Brier score are different estimands. Their exact difference here is the empirical within-cell outcome variance. Positive affine calibration preserves the displayed aggregate winner.

evaluator	mean predicted-real	cell-rate MSE	empirical Brier	intercept	slope	task-held-out recalibrated MSE	winner preserved?
Cosmos	-0.08750	0.02396	0.20417	0.133	0.791	0.02119	yes
IRASim	-0.14583	0.05563	0.23583	0.240	0.402	0.03693	yes

S8.4 Nonlinear calibration and Murphy decomposition

The isotonic map is fitted and evaluated on the same 12 cells. It is an in-sample shape sensitivity, not prospective calibration or selection repair.

evaluator	raw winner	isotonic winner	raw rate MSE	isotonic in-sample MSE	Murphy reliability	resolution	Brier skill vs prevalence
Cosmos	OpenVLA	OpenVLA	0.02396	0.01149	0.01333	0.02082	0.035
IRASim	Octo-Base	OpenVLA	0.05563	0.02131	0.05219	0.02800	-0.114

S9. Exact displayed-task composition sensitivity

Exact sensitivity over every non-empty subset of complete displayed task/condition blocks; not a sampling probability.

case	leave-one-task-out correct	all nonempty subsets correct
WorldGym	17/17	131068/131071
Digital Cousins	4/4	15/15
SIMPLER Google	5/5	24/31
Cosmos-Surg manual	0/4	2/15
WM-PolicyEval / Cosmos	4/4	12/15
WM-PolicyEval / IRASim	1/4	4/15

S10. Checkpoint-selection sensitivity

All tied maxima are enumerated. A range therefore reflects the selection rule itself, not arbitrary file order.

task	checkpoints	all-checkpoint r	best-real r range	tie combinations	best-sim r range	tie combinations
T	15	0.915	0.878 to 0.944	2	0.878 to 0.947	2
rope	20	0.901	-0.471 to -0.103	4	-0.683 to -0.683	1
sloth	17	0.944	0.905 to 0.941	4	0.933 to 0.968	2

S11. MMRV versus correlation stability

MMRV uses SIMPLER’s strict-ordering XOR, real-side gap, and divide-by-N convention. RoboWorld’s real-side leaderboard score is min-max scaled once on the full panel.

dataset	k	MMRV	absolute MMRV range	relative r swing	relative MMRV swing	ratio
Digital Cousins, by policy	4	0.111459	0.020831	1.9%	18.7%	9.8×
Digital Cousins, by generalization level	4	0.111459	0.088890	8.1%	79.8%	9.8×
RoboWorld 10a	8	0.005765	0.003467	15.4%	60.2%	3.9×
RoboWorld 10b	8	0.055067	0.044383	30.0%	80.6%	2.7×

S12. Real2Sim MMRV convention reproduction

The declared grid has 60 named entries but 48 distinct formulas because one predicate pair is pointwise identical by construction. Exactly one distinct formula matches all three Table I values and all three Figure 9 values: less-than-or-equal XOR (equivalently strict-> XOR), simulated-side gap, divide by N.

panel	N	episodes	recovered r	printed r	exact recovered MMRV	printed MMRV
toy packing	17	20	0.9444	0.944	13/170 (0.076471)	0.076
rope routing	20	27	0.9007	0.901	47/270 (0.174074)	0.174
T-block pushing	15	16	0.9147	0.915	13/120 (0.108333)	0.108

The following appendix values use the same recovered convention:

panel	exact recovered MMRV	decimal
Figure 9 toy_packing	21/200	0.105000
Figure 9 rope_routing	307/2000	0.153500
Figure 9 t_block_pushing	209/3000	0.069667

We recover 15 T-block checkpoints from Figure 3; they reproduce Table I’s $r = .915$ and $\text{MMRV} = .108$. Figure 10 reports a separate 12-checkpoint replay subset.

S13. Model-conditional correlation sensitivity

These are equal-tailed posterior intervals under iid bivariate-normal unit summaries and a uniform prior on ρ . They are not assumption-free intervals.

task	unit-level r	n	95% posterior interval
T	0.962	4	[-0.256, 0.978]
rope	0.697	4	[-0.538, 0.901]
sloth	0.984	4	[-0.163, 0.989]